

Predicting College Acceptance

By: Jason, Jonah, Rehan

Two teal lines, one slightly above the other, that start from the left and curve towards the right, ending under the authors' names.

Topic and Dataset

What are the primary factors that affect college acceptance?

We chose college acceptance because we were interested in what factors most strongly determine whether someone gets admitted into college or not and we wanted to see if we could predict whether they got admitted.

Dataset:

<https://github.com/shreybirmiwal/college-predictor/blob/main/1-DataPrep/7-manually-clean/CLEAN.csv>

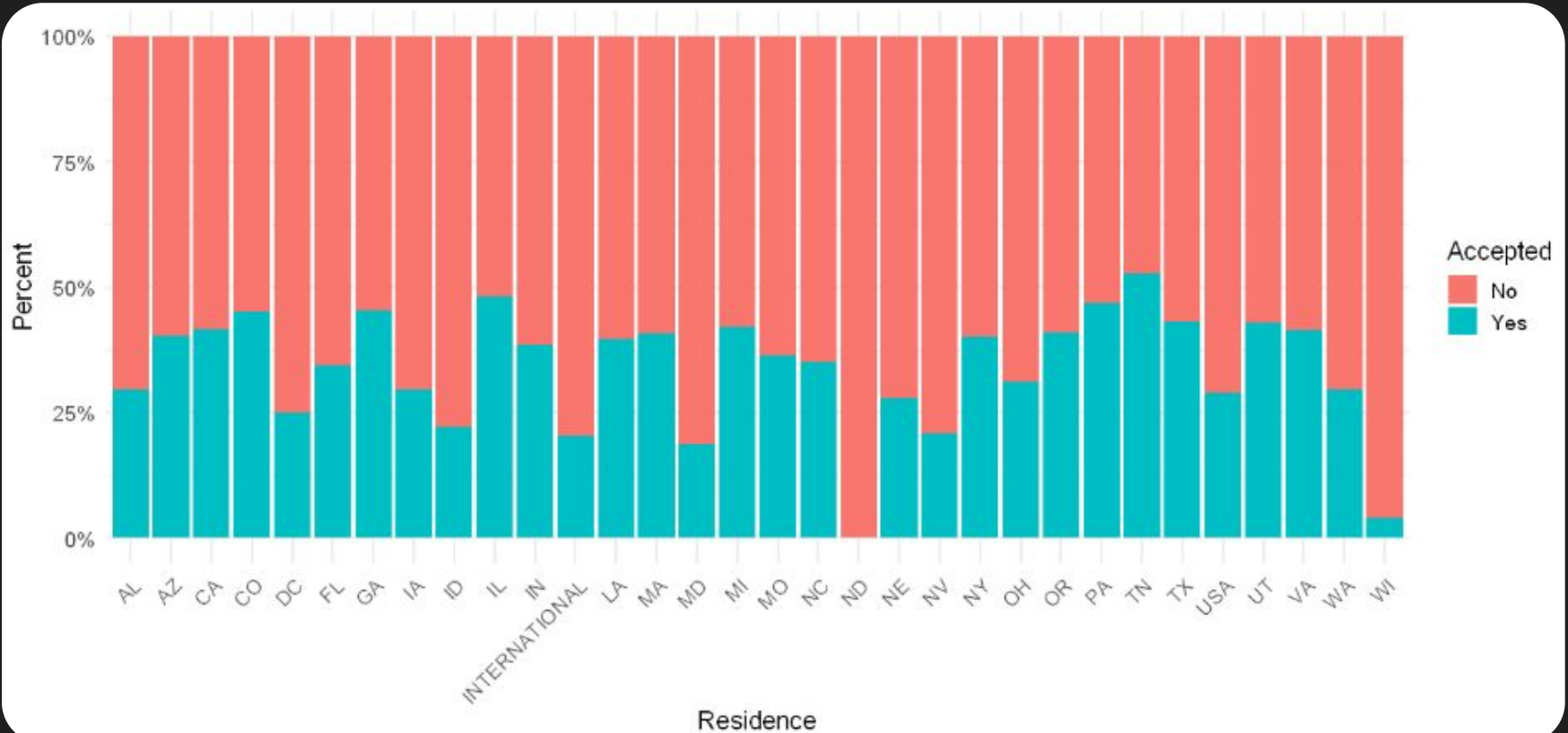
About the Dataset

ARIZONA STATE	BERKELEY	BOSTON COLLEGE
71	275	36
BOSTON UNIVERSITY	BROWN	CALTECH
95	256	115
CASE WESTERN	CLAREMONT MCKENNA	CMU
85	13	152
COLGATE	COLUMBIA	CORNELL
28	275	320
DARTMOUTH	DUKE	EMORY
145	217	88
FORDHAM	GEORGIA TECH	HARVARD
79	166	293
HARVEY MUDD	JOHNS HOPKINS	MIT
27	124	237
NORTHEASTERN	NORTHWESTERN	NYU
219	231	167
POMONA	PRINCETON	PURDUE
70	254	161
RICE	RUTGERS	San Jose State University
117	42	39
STANFORD	SWARTHMORE	TAMU
330	67	41
TUFTS	TULANE	UCD
116	35	105
UCHICAGO	UCI	UCLA
93	133	249
UCM	UCR	UCSB
17	50	113
UCSC	UCSD	UIUC
65	153	179
UMICH	UNC CHAPEL HILL	UPENN
201	57	355
USC	UT-AUSTIN	UT-DALLAS
206	113	19
UVA	UW MADISON	VANDERBILT
139	45	118
VIRGINIA TECH	WASHU	WILLIAMS
39	97	73
YALE		
302		

This dataset captures anonymized demographic profiles (gender, race, U.S. residence, and income tier) and application criteria (high school type, institutional hooks, intended major, SAT scores, and normalized ratings for extracurriculars, essays, recommendations, and interviews) for 1,485 prospective college students alongside their final application outcomes across major U.S. universities.

To clean the data we combined the waitlist and rejected columns and also combined all the other genders into LGBTQ+ to keep the data simple. In the raw dataset, each row represented a single student, and their college results were grouped together as lists inside the 'accepted', 'waitlisted', and 'rejected' columns. We restructured the data so that every college application gets its own separate row. If a student applied to three colleges, they now have three individual rows showing a clear acceptance or rejection for each school.

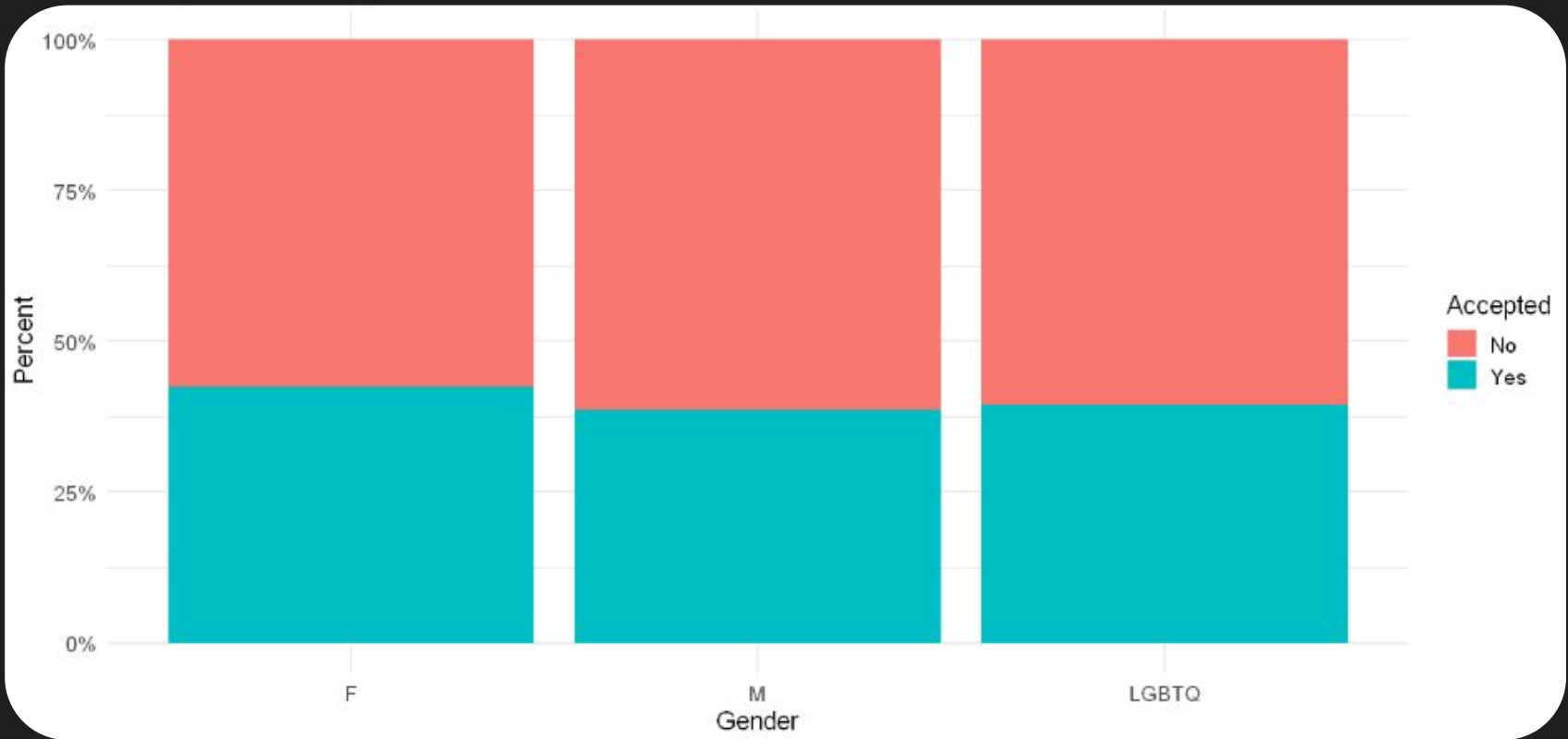
Acceptance Rate by Residency



Greatest proportion of acceptance - Tennessee

Least proportion of acceptance - North Dakota

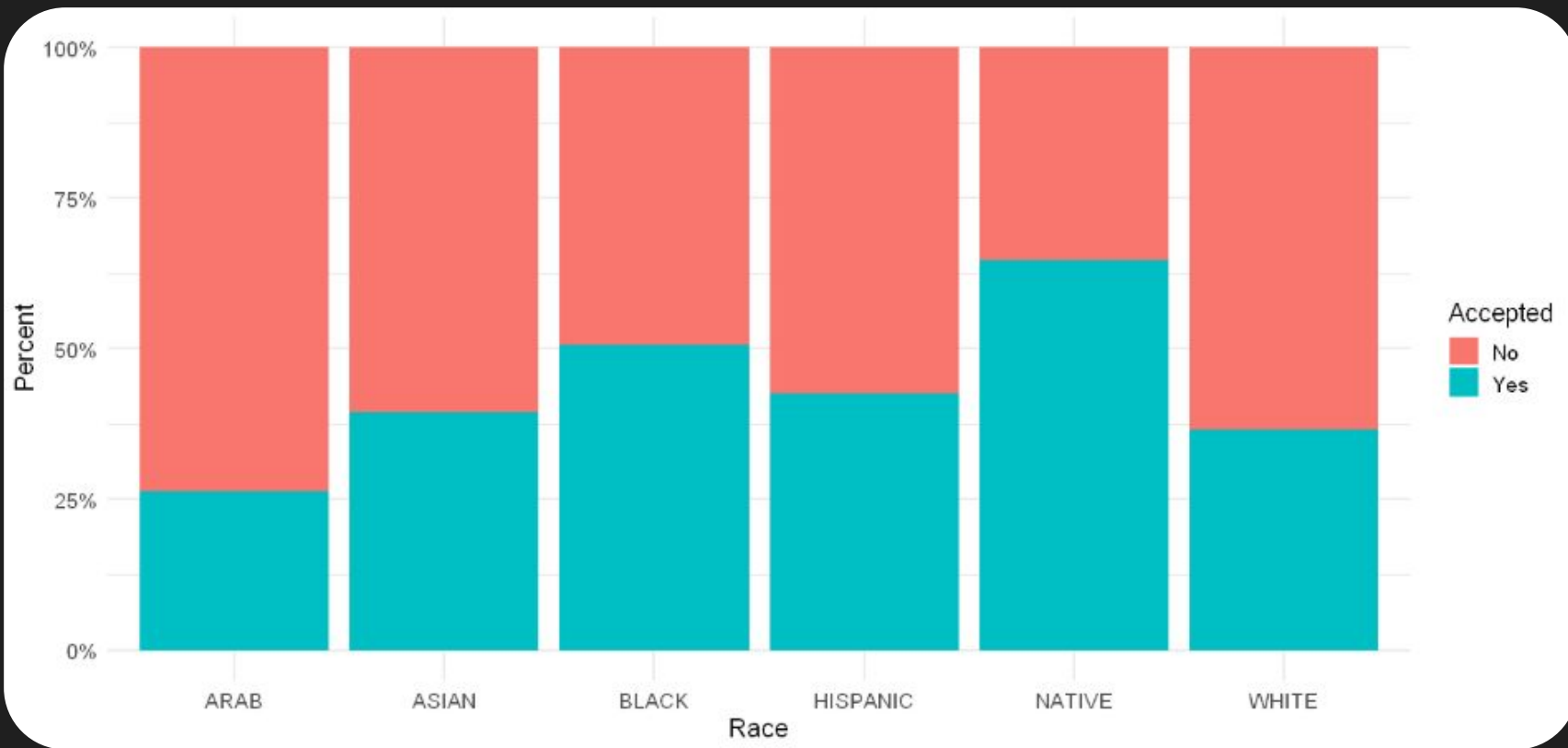
Acceptance Rate by Gender



Greatest proportion of acceptance - Female

Least proportion of acceptance - Male

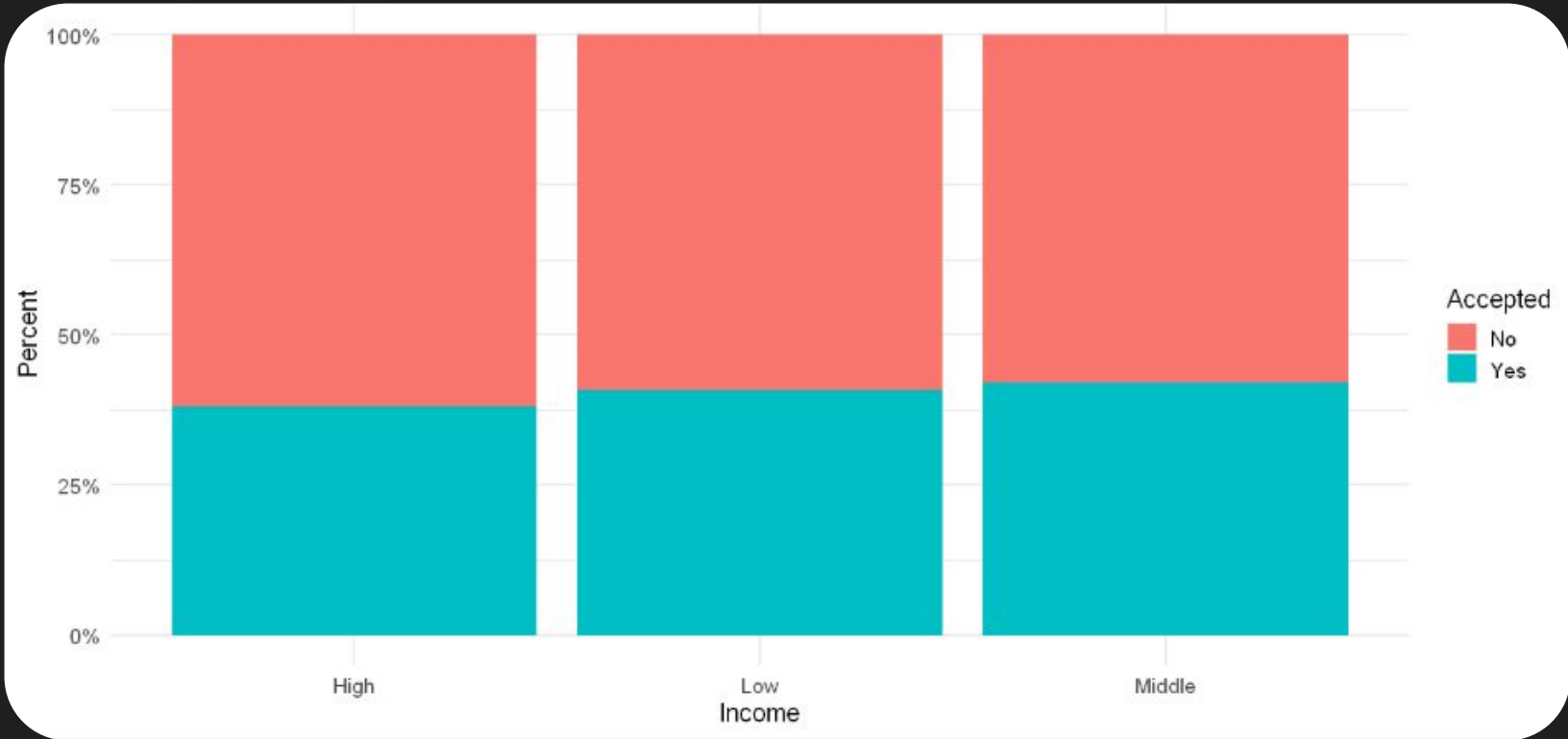
Acceptance Rate by Race



Greatest proportion of acceptance - Native

Least proportion of acceptance - Arab

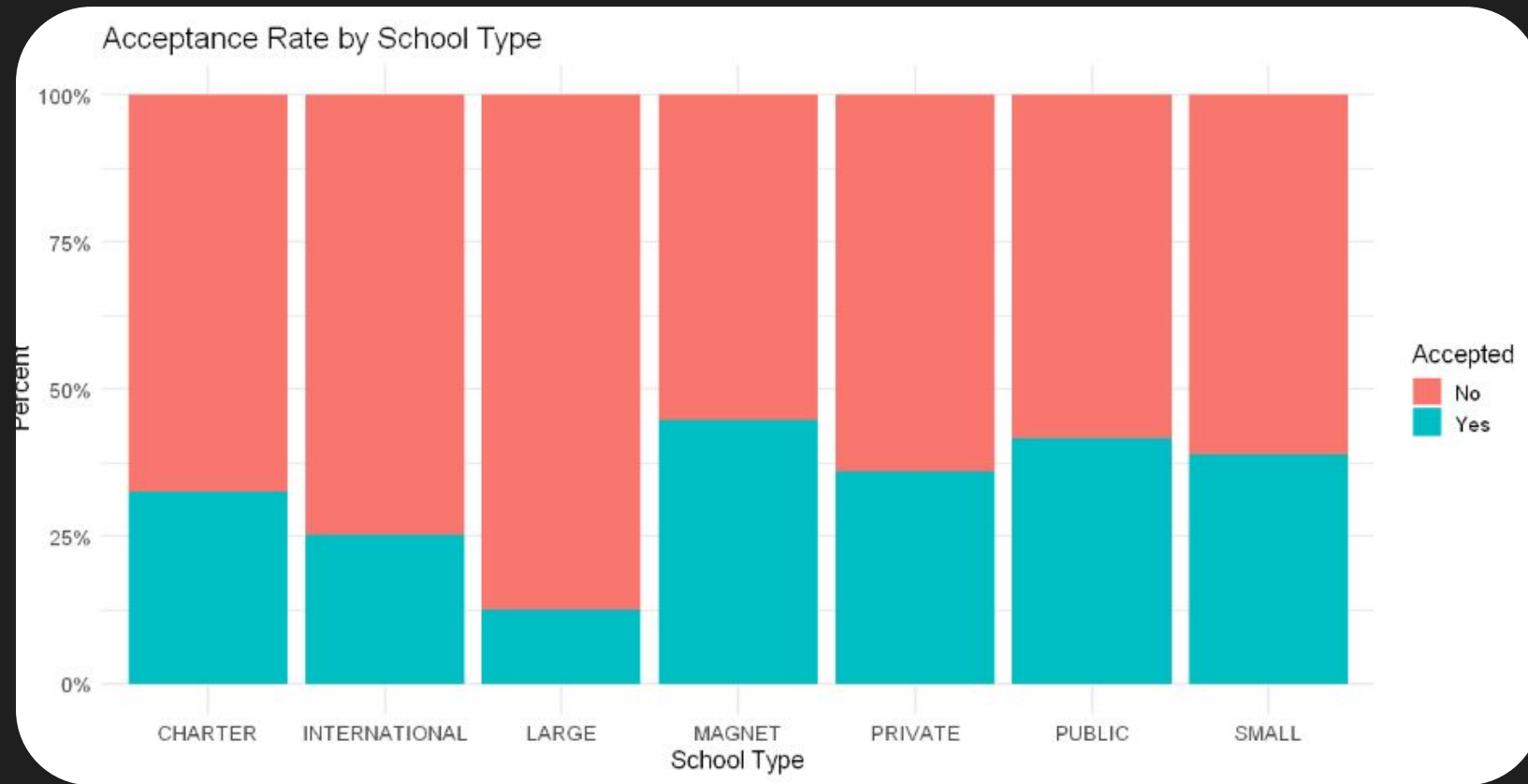
Acceptance Rate by Income



Greatest proportion of acceptance - Middle

Least proportion of acceptance - High

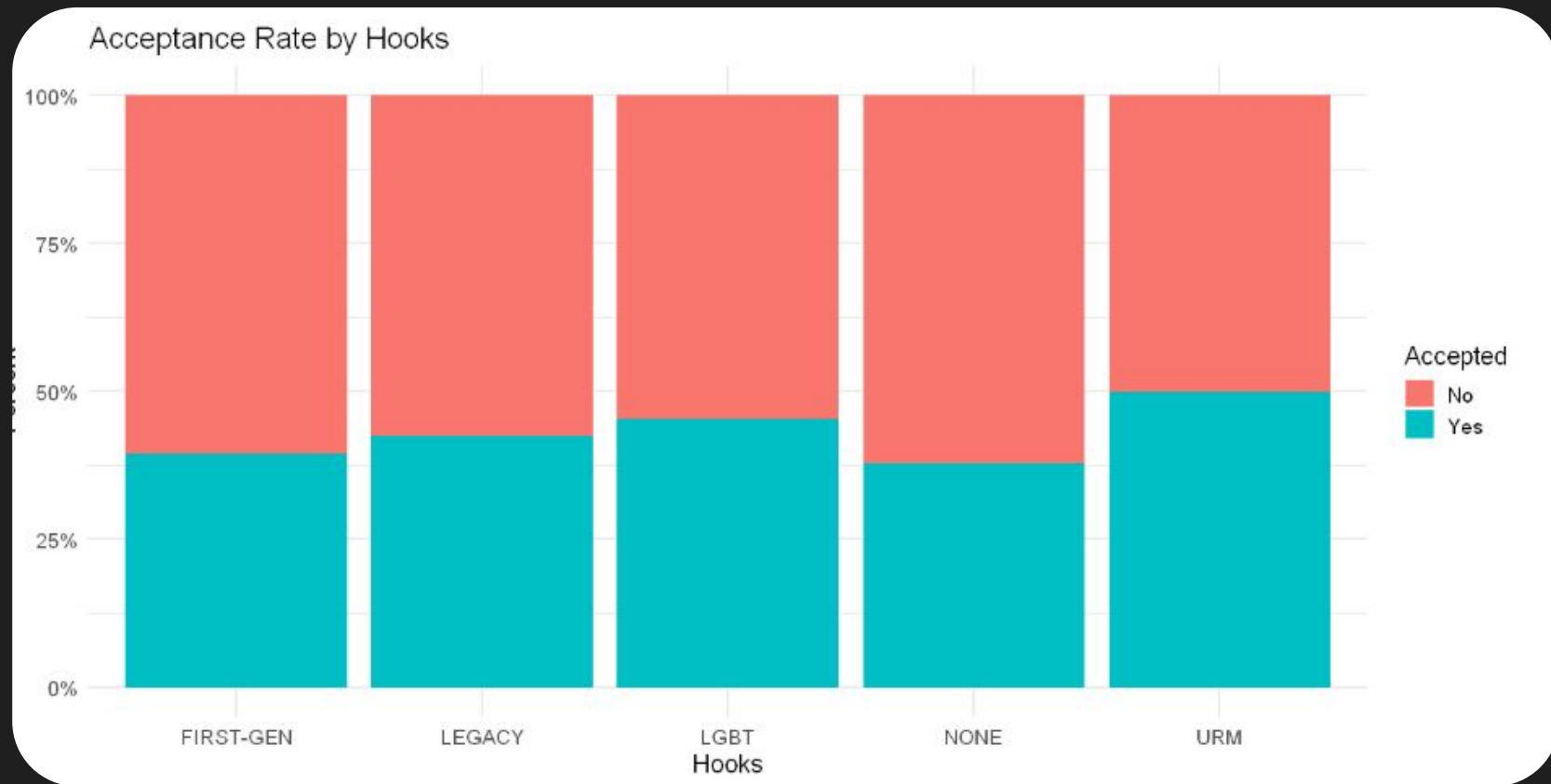
Acceptance Rate by School Type



Greatest proportion of acceptance - Magnet

Least proportion of acceptance - Large

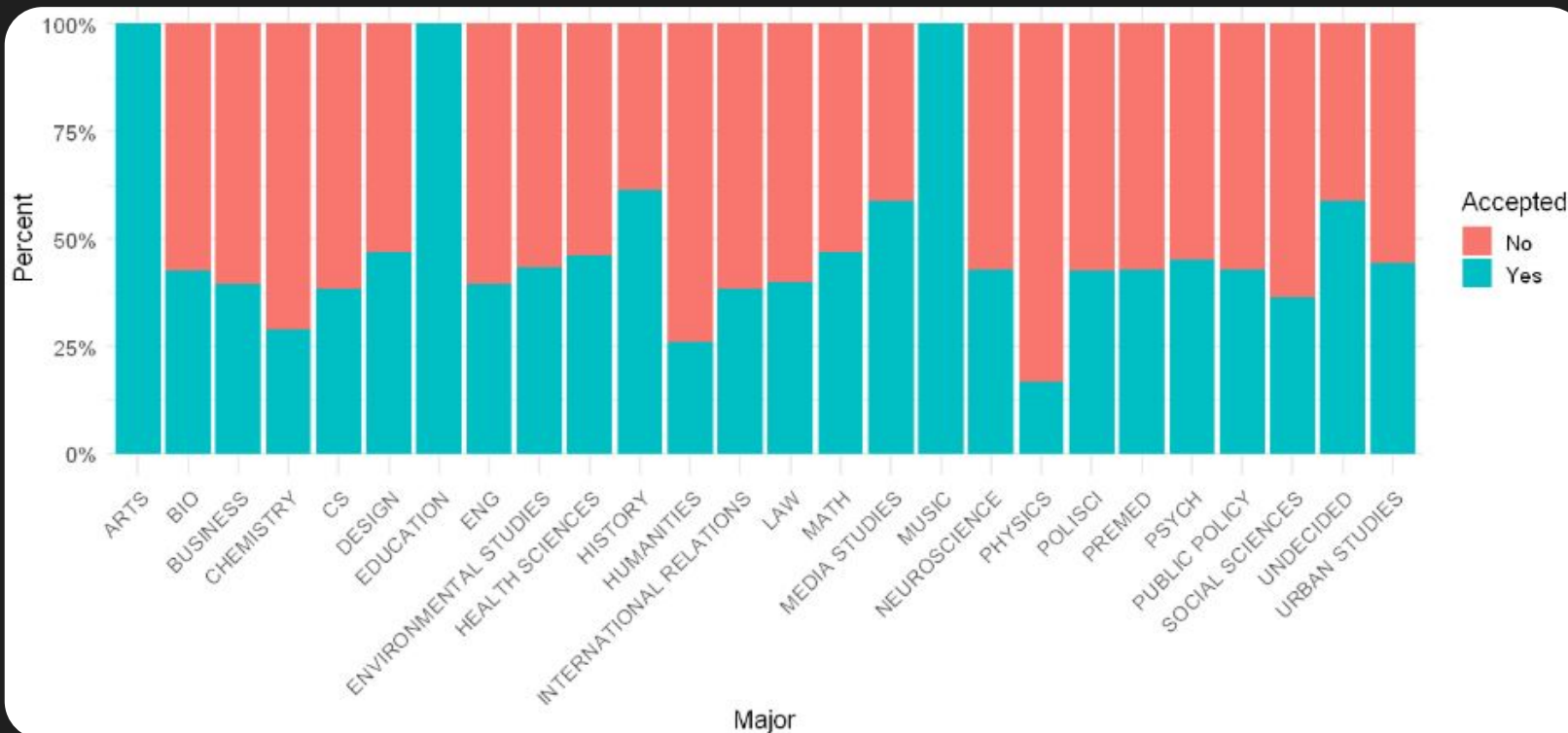
Acceptance Rate by Hooks



Greatest proportion of acceptance - URM

Least proportion of acceptance - None

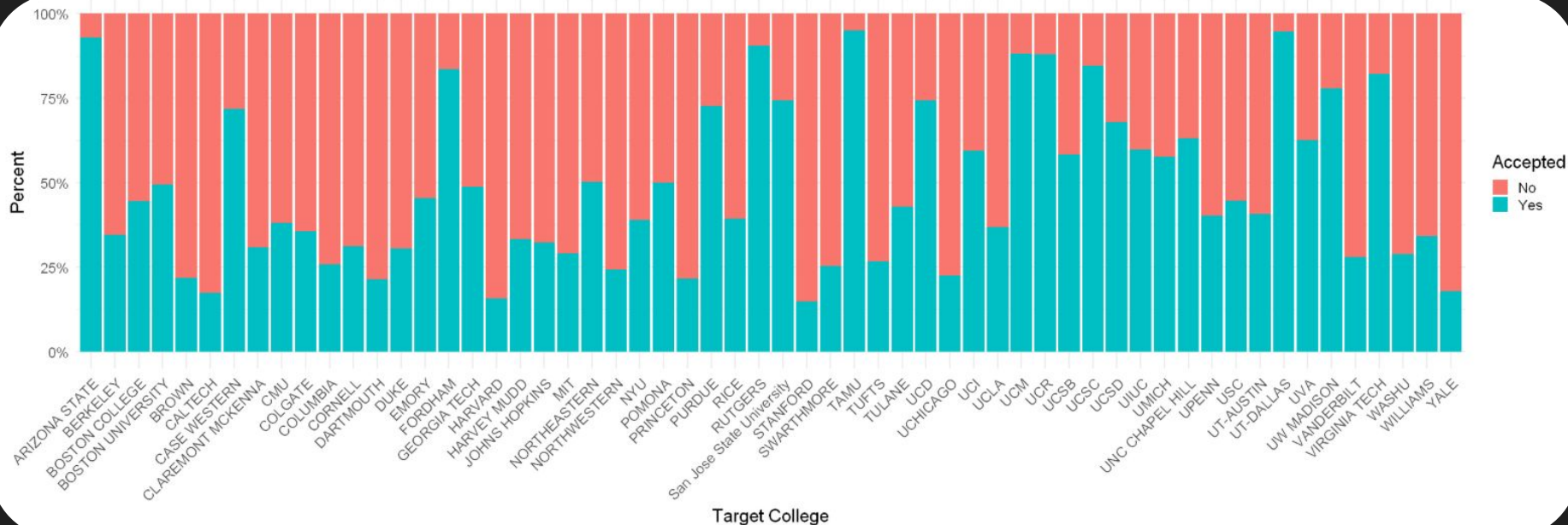
Acceptance Rate by Major



Greatest proportion of acceptance - Music, Education, Art

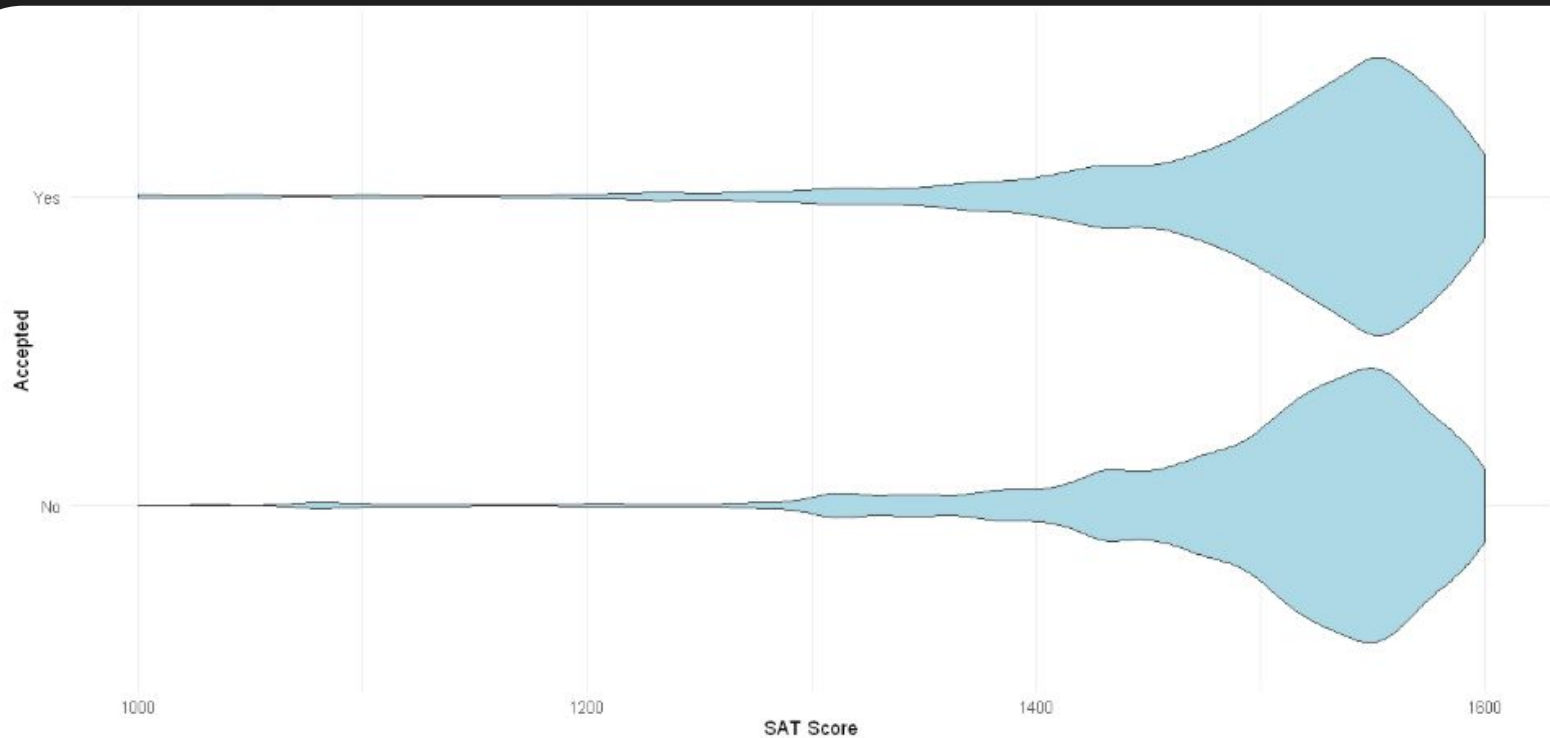
Least proportion of acceptance - Physics

Acceptance Rate by Target College



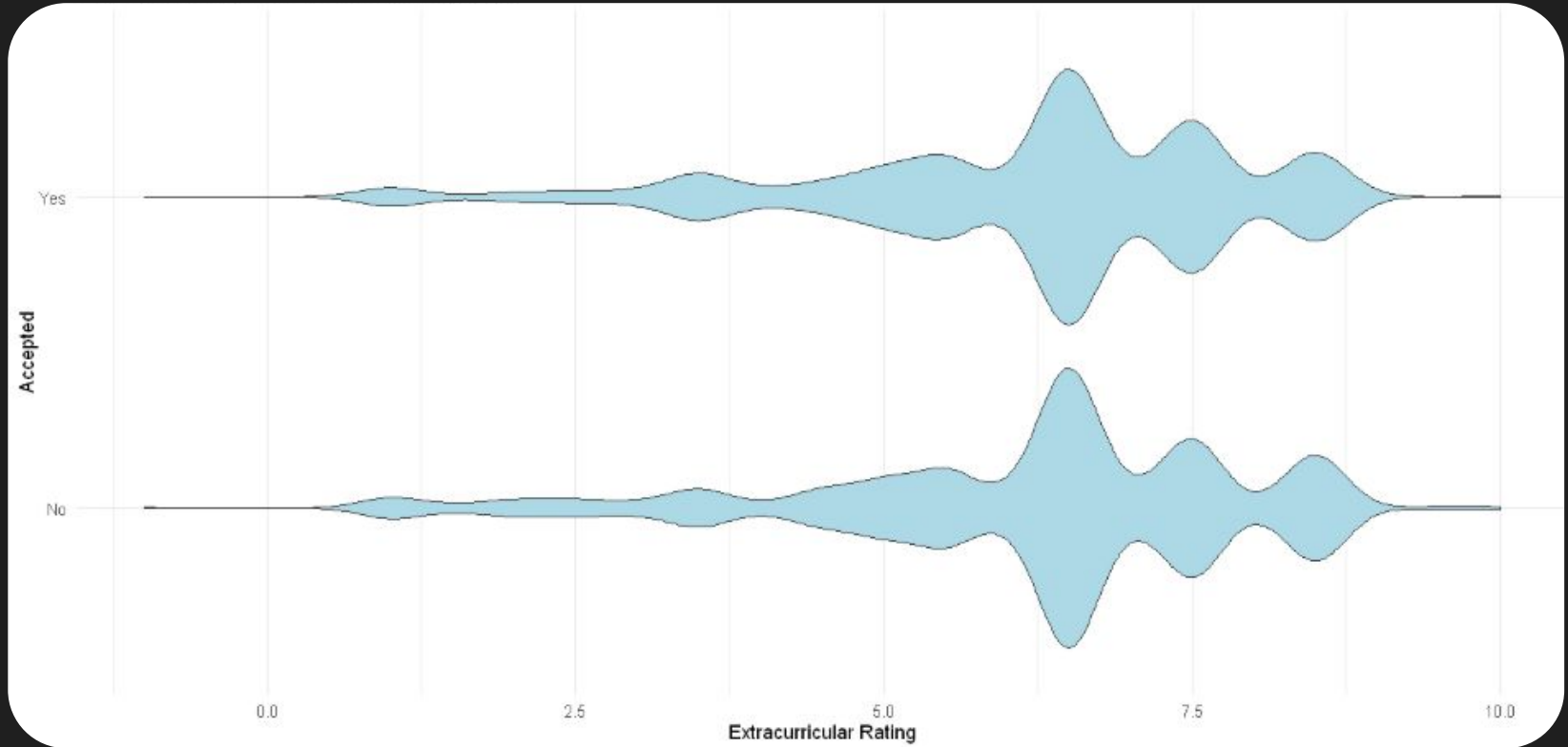
Greatest proportion of acceptance - Tamu Least proportion of acceptance - Stanford

Acceptance Rate by SAT



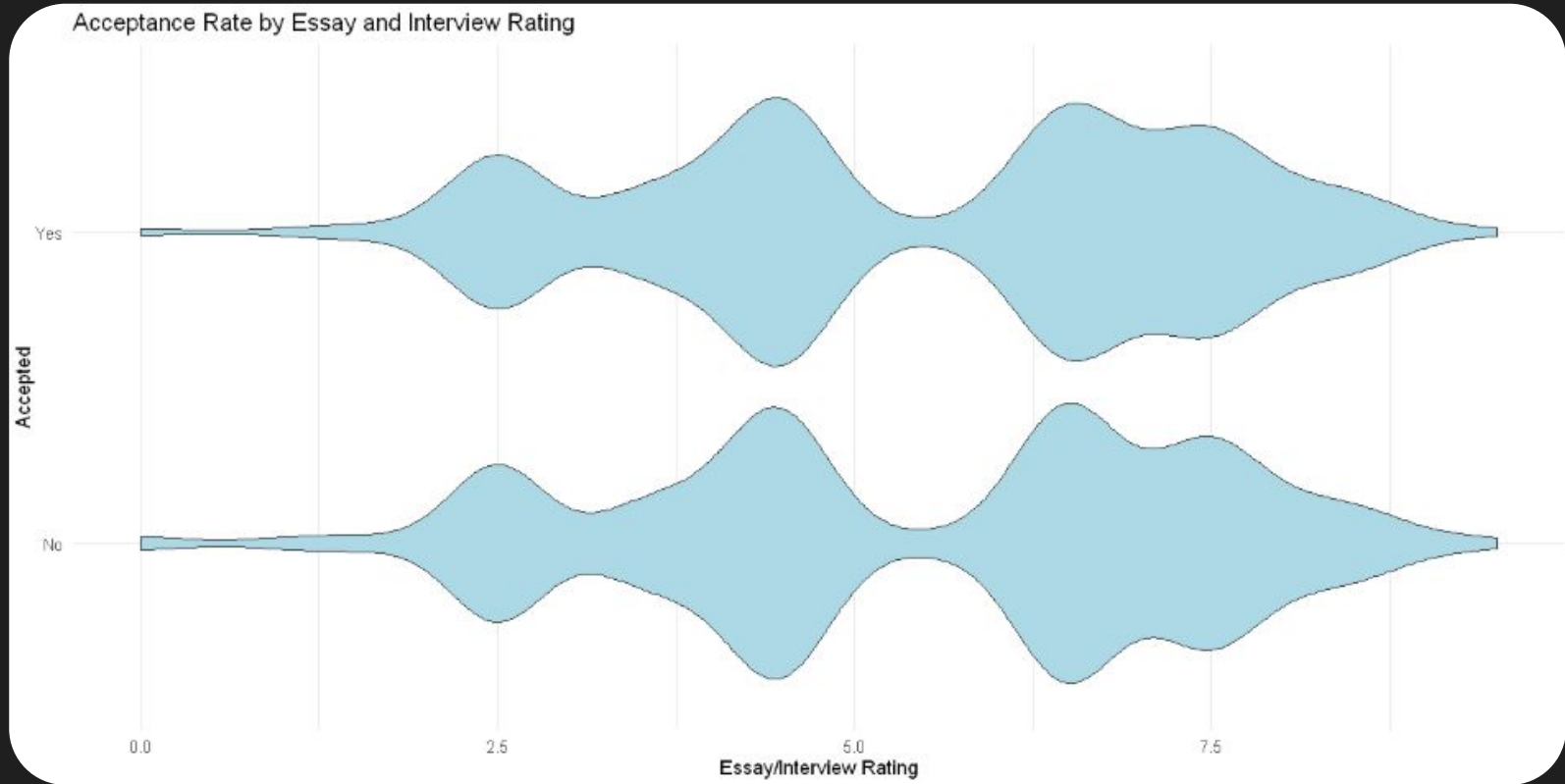
The accepted group and the rejected group had almost identical SAT score distributions.
They both peaked at ~1550

Acceptance Rate by EC Rating



The accepted group and the rejected group had almost identical EC rating distributions. They both peaked at ~6.2

Acceptance Rate by Essay/Interview Rating



The accepted group and the rejected group had almost identical Essay/Interview rating distributions. They both peaked at ~4.6 and ~6.2

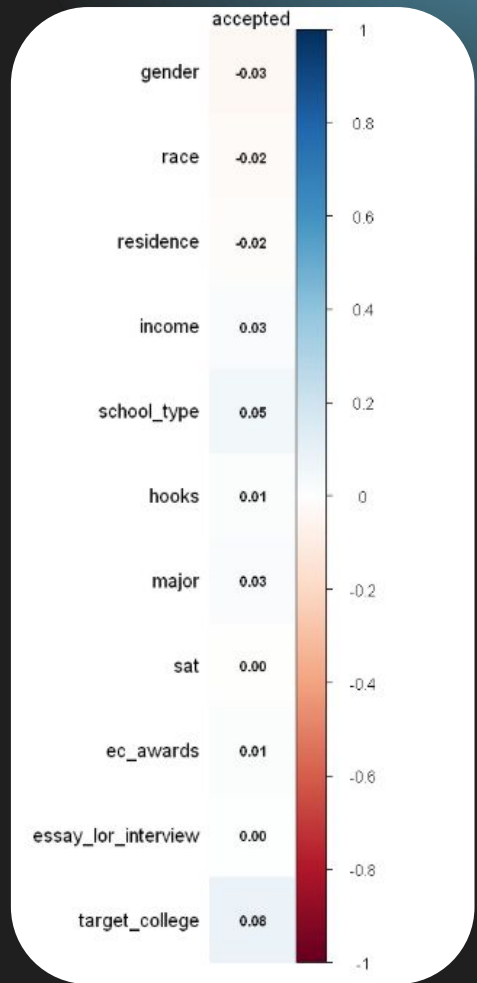
Correlation

None of the variables had a strong correlation with college acceptance.

The variable with the strongest correlation was target college because the college you apply to greatly affects your chances of getting accepted.

A variable like SAT *should* have a strong correlation, but the correlation with acceptance is 0. This can be explained for 2 reasons:

- The average SAT score for both accepted and rejected was 1500. However correlation requires variance to make a trend.
- The SAT is a *threshold*. Getting a 1560 does not make you more likely to be accepted than a 1550. However, getting a 1512 heavily beats a 1480 (when applying for Purdue).

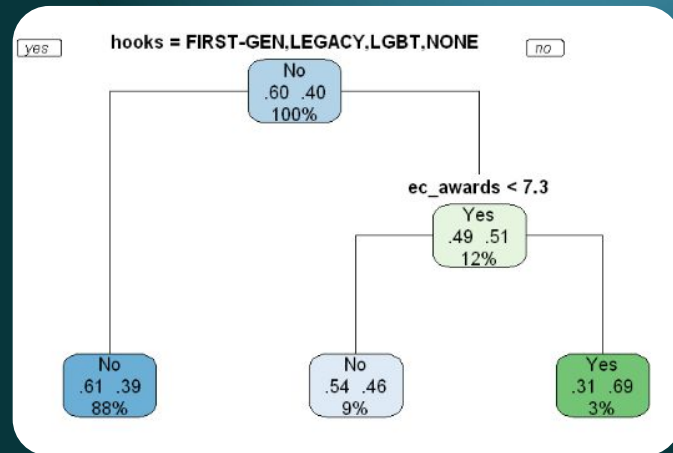


Random Forest Explained

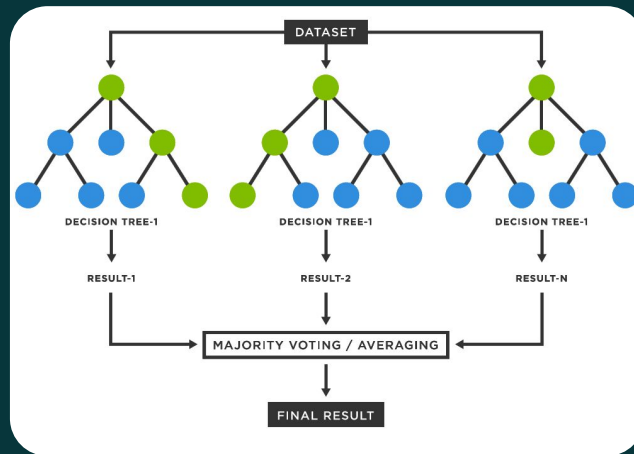
A random forest is a machine learning model that makes predictions by combining the outputs of multiple decision trees.

- A decision tree behaves like a flowchart with yes/no questions.
- A single tree memorizes training data rather than learning general patterns, making it inaccurate with new data.

A random forest creates random subsets of the data, and out of those subsets, takes random columns from the subsets to make multiple decision trees. Once the decision trees are trained, the model makes a decision by choosing the answer that the majority of the trees got.

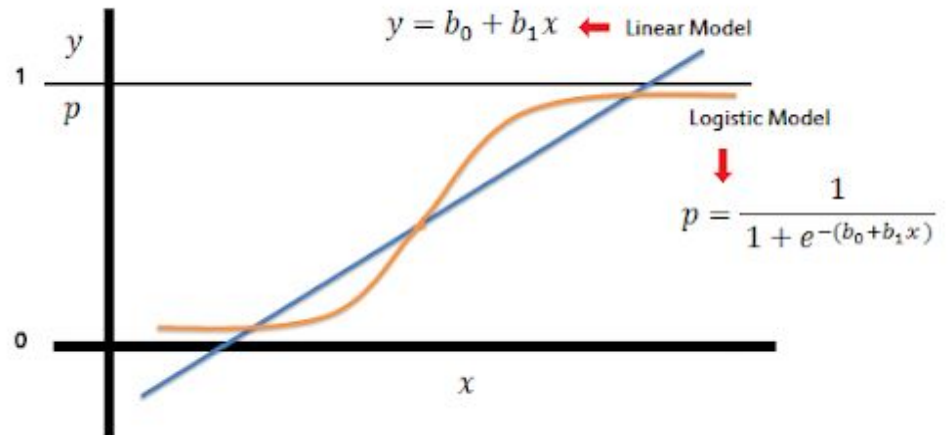


Random Forest Visualized



Logistic Regression Explained

Logistic regression is a classification algorithm used to predict the probability of a binary, "yes-or-no" outcome. Because college admission is strictly a two-choice result (admitted or rejected), standard linear regression fails because it draws a straight line that outputs impossible probabilities below 0% or above 100%. Choosing logistic regression applies an S-shaped curve that forces all predictions to fall accurately between a 0% and 100% probability based on your student statistics.



Random Forest Example

gender	race	residence	income	school_type	hooks	major	sat	ec_awards	essay_lor_interview	target_college	answer	prediction
<fct>	<fct>	<fct>	<fct>	<fct>	<fct>	<fct>	<int>	<dbl>	<dbl>	<fct>	<fct>	<fct>
M	HISPANIC	CA	Low	MAGNET	URM	CS	1520	6.5	3.5	UPENN	Yes	Yes
M	ASIAN	OR	High	PUBLIC	NONE	BIO	1410	5.5	7.5	HARVARD	No	No
M	WHITE	MI	Low	PUBLIC	URM	CS	1450	2.5	4.2	STANFORD	No	Yes

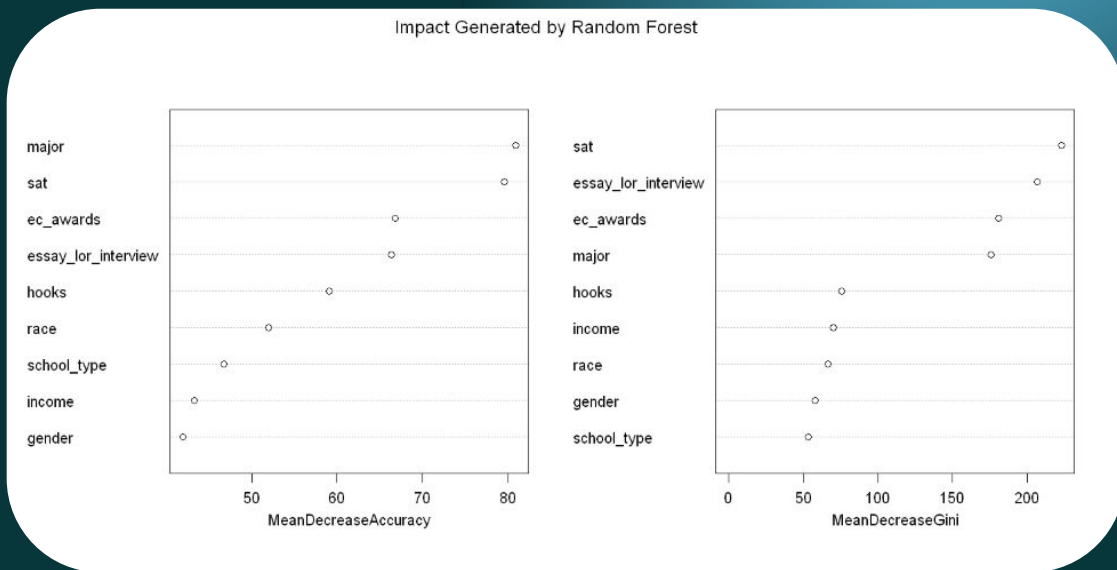
Logistic Regression Example

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<fct>	<fct>	<fct>	<fct>	<fct>	<fct>	<fct>	<int>	<dbl>	<dbl>	<fct>	<fct>	<fct>
M	HISPANIC	CA	Low	MAGNET	URM	CS	1520	6.5	3.5	UPENN	Yes	No
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M	WHITE	MI	Low	PUBLIC	URM	CS	1450	2.5	4.2	STANFORD	No	No

Impact of each variable

These two graphs show what matters most when predicting college acceptance, looked at in two different ways. The left graph shows that removing the major variable causes the biggest drop in overall guessing accuracy, making it the most critical factor for final correct predictions.

The right graph shows that sat scores are the best at cleanly sorting and splitting the applicant data into separate groups. In short, while SAT scores are great for broadly organizing students, a student's intended major is what ultimately drives the model's final, accurate decisions. The top 5 most impactful things are the same for both graphs so they are likely good predictors of college acceptance. We did not include residency in impact or in our prediction model because there were some very underrepresented states which skewed the impact and our predictions a lot. The difference between impact and correlation is that impact is how these factors directly affect your chances of getting into college and correlation is the linear relationship between two variables. This is why the factors at the top of the impact list are at the bottom for correlation.



Confusion Matrices

The Confusion Matrices show the results of the models, showing true positives, false positives, true negatives, and false negatives.

Both models contain more “No”s than “Yes”s for 2 main reasons:

- 60% of applicants were rejections. This makes the models start with a bias towards “No”
- Classification models default to a cutoff of 50% probability. In competitive college admissions, a flawless application might have a 35-45% chance of getting accepted (which is really good for Stanford’s 4% acceptance rate). However, it still falls below the cutoff, making it automatically a rejection.

Logistic Regression

Prediction	Reference	
	No	Yes
No	887	549
Yes	62	83

Random Forest

Prediction	Reference	
	No	Yes
No	757	346
Yes	192	286

Comparison of the 2 Models

The accuracy of the Random Forest was 5% higher than Logistic Regression. RMSE is a value that measures the average difference between the model and the actual data. This means a value closer to 0 is better. The Random Forest beat Logistic Regression by 0.04 points.

This is expected because:

- Logistic Regression is rigid, Random Forest is flexible
- Logistic Regression handles one variable at a time, Random Forest can handle multiple

Model	RMSE	Accuracy
Random Forest	0.40	66%
Logistic Regression	0.44	61%

Conclusion

We concluded that our most successful model was the random forest model, and we found that the strongest predictors of college admission were college choice, major, SAT, extracurriculars, hooks, and a combined qualitative category for personal application materials.

One limitation we encountered was not enough data. We had some underrepresented categories where because there is so little data the results are very skewed one way which affects our predictions a lot. This would not be a problem if we had a much bigger dataset because then we would have enough data in the underrepresented categories and the results would not be as skewed. Additionally, using the 50% probability threshold caused numerous false negatives for students applying to low-acceptance rate colleges. This could be fixed by lowering the threshold to ~35-45%

One practical implication of these models is that they shift college classification to be more specific than Safety/Match/Reach. These models replace guesswork with an accurate probability distribution, making the recommendations highly personalized.

Overall, while these models have their drawbacks, their overall implications and future uses can greatly improve the quality of life for both colleges and students from all backgrounds.



Thank You